

Arbaz Siddiqui

Principal Software Engineer · AI Infrastructure and Distributed Systems

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SUMMARY

Backend engineer, ten years, mostly consumer products at scale. At Turnip I built and ran the platform behind 25M users, then spent two years on the AI infrastructure around it. Outside that I ship my own work, an open 12B language model, two live apps, and content pipelines at 500K+ monthly views.

EXPERIENCE

Principal Software Engineer · Turnip

Jan 2021 – Present

Four pivots in one startup. Game streaming, live audio, a creator social network, now Zaps, an AI design suite.

- Built and ran the backend behind 25M users. 20+ services on Kubernetes, written in Node.js and Go, fronted by Kong with Kafka and Redis carrying traffic.
- Built RTMP and FFmpeg streaming at 50K streams a day, sub 5s latency. Audio rooms with MQTT holding state for 2M monthly users over 70M call minutes. A social network with connections on Neo4j, indexed on Elasticsearch.
- Shipped AI photo editing on ComfyUI, orchestrated with LangGraph and n8n. Moving generation to a distributed GPU network cut GPU spend 80% at sub 2s an image, and moving to serverless took 70% off infrastructure.
- Replaced template tag search with semantic search over embeddings, retrieved by vector similarity.

Senior Backend Engineer · MouseBelt

Apr 2020 – Jan 2021

- Built a centralized exchange. Accounts, KYC, markets and order flow, a matching engine in C with gRPC and WebSocket layers, custody across Bitcoin, Ethereum, ICON and BNB Chain, plus a DEX aggregator and indexer.

Lead Backend Engineer · Paytm Insider

2017 – Apr 2020

- Led a team of seven on the movie ticketing platform, which held 500+ requests a second on sale nights, and designed the logistics architecture behind delivery opt-in, order tracking and shipment tooling.
- Built the commerce flow of cart, payments, inventory and taxation, run as AWS microservices on Kubernetes with Lambda and Kinesis carrying the on-demand work.

SELECTED PROJECTS

Ozan-v1-12B · hf.co/arbazsiddiqui. Trained and released with QLoRA SFT and anti-slop DPO on one A100 for about \$40. Lowest slop of any 12B on EQ-Bench, below Gemma 3 27B. #1 trending on Hugging Face, 3K+ downloads.

The Footnote · @thefootnotemedia. An autonomous pipeline that researches, scripts, voices, illustrates, renders and publishes at 500K monthly views. The same growth pipeline drives a horror channel and Iris's App Store traffic.

Open source · github.com/arbazsiddiqui. A decade of public code, 300+ stars across it. lru-cache-node, an O(1) LRU cache with expiry featured in Node Weekly, and anabranh, an HTTP load balancer and reverse proxy in Go.

Iris and Mercuro · [App Store](https://AppStore). A privacy-first cycle tracker at 4.9 stars in twelve languages, and a logic puzzle game with its own constraint solver and generator, on iOS and Android.

SKILLS

AI: fine-tuning, RAG (LangChain, embeddings, pgvector), diffusion (ComfyUI, Flux, Qwen, Wan 2.2), agent orchestration (LangGraph, MCP servers, Claude Agent SDK), workflow automation (n8n), llama.cpp, distributed GPU inference

Backend: Go, Node.js and TypeScript, Python · Kafka, Redis, MQTT, gRPC, WebSockets, RTMP and FFmpeg, Postgres, MongoDB, Elasticsearch, Neo4j, Cassandra · event-driven and hexagonal architecture

Infrastructure: Kubernetes (GKE), Kong, Docker, GCP, AWS, serverless (Lambda, Kinesis, SQS), CI/CD, Prometheus, ELK

Crypto: order matching engines, multi-chain custody, DEX aggregation, NFT systems